PAVEAISTRUCT™ | PAVEMENT DESIGN & VOLUMETRIC CALCULATION NOTE

DOC. REF: NW-B-CAD-1179

Mechanistic-Empirical Pavement Analysis Suite • IRC:37-2018 / AASHTO Framework

GENERATED: Oct 13, 2026 11:00 PM

Active Section: NH-48/PKG-04/CH:12+400-13+400 • Model Kernel: Multi-Output RF v2.4

RELIABILITY: 90% (Expressway Standard)

REGULATORY COMPLIANCE ASSESSMENT: **IRC:37 COMPLIANT (PASS)** Design Traffic: 133.0 MSA | Carriageway Area: 7,000 m²

1. STRUCTURAL LAYER ARCHITECTURE & INPUT PROPERTIES

Pavement Component	Dimension / Value	Mechanical Properties	Specification Standard
Design Traffic Loading	133.0 MSA	Standard 80 kN Dual-Tire Tandem Axle	IRC:37-2018 Spectrum (90% Rel)
Layer 1: Bituminous Course	195 mm	Dynamic Modulus E• = 3,0	BC Grade-1 + DBM (VG-40 Bitumen @ 35°C)
Layer 2: Unbound Granular Base	250 mm	Equivalent Modulus E, = 2	Wet Mix Macadam (WMM) + GSB Grade-I
Layer 3: Subgrade Soil Medium	Semi-Infinite	Resilient Modulus Ef = 50	Design CBR = 5% (MR = 17.6 × CBR^0.64)
Project Corridor Geometry	Length: 1.0 km	Pavement Width: 7.0 m	Footprint: 7,000 m² (2.5% Camber)

2. MECHANISTIC STRAIN RESPONSES & FAILURE CRITERIA

Mechanic Response Parameter	Surrogate Value	Permissible Limit	Safety Margin	Cumulative Damage	Assessment
Critical Tensile Stra	148.7 με	"d 207.2 με	+28.3% (Margin)	Damage DF = 0.78	**PASS (Optimal)**
Critical Compressive	435.4 με	"d 551.3 με	+21.0% (Margin)	Damage DR = 1.10	**PASS (Safe)**
Fatigue Repetition Capacity (Nf)	3.46e+41	1.00e+35 Min Threshold	Nf = 171.04 Million Repetitions	IRC:37-2018 Transfer Eq	**VERIFIED**
Rutting Capacity (Nr)	1.59e-16	1.00e-14 Max Threshold	Nr = 120.88 Million Repetitions	AASHTO Subgrade Rut Eq	**VERIFIED**

3. BILL OF QUANTITIES (BoQ) & VOLUMETRIC SCHEDULE

Material Layer Description	Thickness	Volume (m³)	Compacted Density	Net Mass (MT)	Logistics (20 MT Dumpers)
Bituminous Courses (BC + DBM)	195 mm	1,365.00 m³	2.40 t/m³	3,276.00 MT	"H 164 Dumpers (20 MT)
Granular Base & Subbase (WMM + GSB)	250 mm	1,750.00 m³	2.10 t/m³	3,675.00 MT	"H 184 Dumpers (20 MT)
TOTAL PAVEMENT STRUCTURE	445 mm	3,115.00 m³	Composite	6,951.00 MT	"H 348 Total Fleet Trips

ENGINEERING CERTIFICATION & PROFESSIONAL VERIFICATION

I hereby certify that the mechanistic responses and material takeoff scheduled herein comply with the specifications prescribed in IRC:37-2018 utilizing the verified Multi-Output Continuum Surrogate Kernel.

IRC:37-2018 CERTIFIED

Designed By: Pavement Design Engineer

Approved By: Lead Highway Consultant